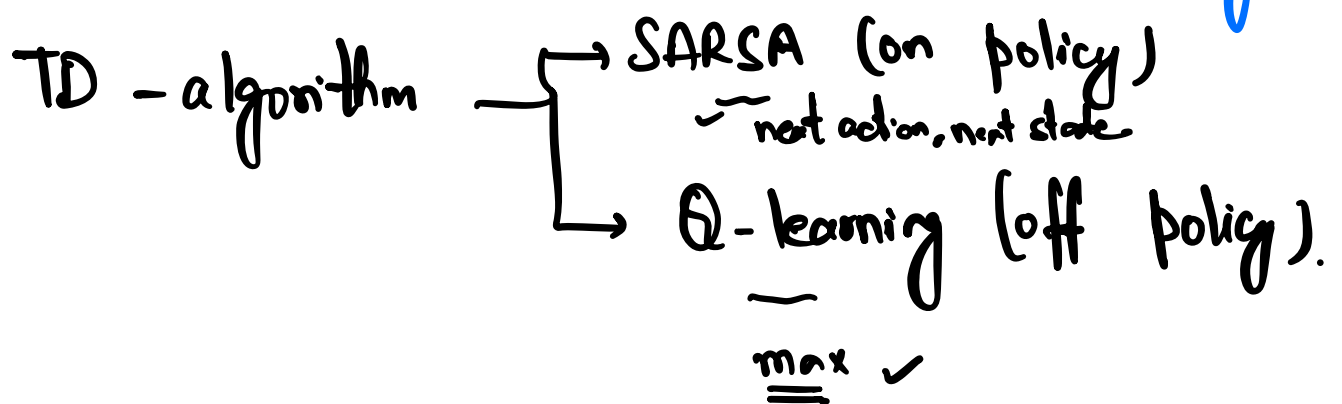
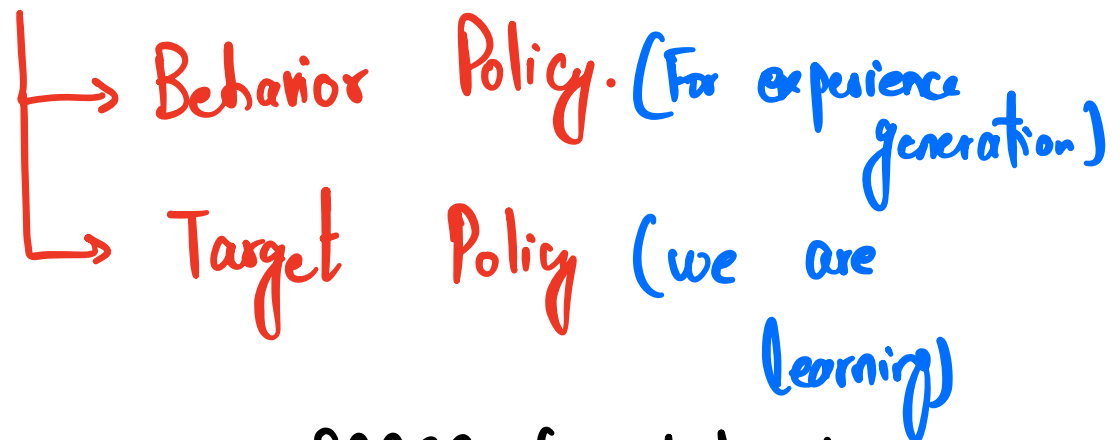


Reinforcement learning..

04/04/2026

Off policy learning..



Monte Carlo Algorithm

→ Law of large numbers.

$$\underline{V}_{\pi} = E[G_t | S_t = s].$$

$V_{\pi} = \text{average of returns}$

} on policy learning in MC

off policy $\rightarrow$ Behavior Policy. $(b) (s_t, A_t)$
 $\rightarrow$ Target Policy. (π)

(s_t, A_t)

(i) Assumption of Coverage:

$$\Rightarrow \underbrace{V(s)}_{\uparrow} = \underbrace{E[G_t | s_t = s]}_{\uparrow}$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 π $?$ (B)

Target $\{ \underline{s}_1 \rightarrow \begin{matrix} \nearrow A_4 (0.2) \\ \rightarrow \underline{A_1} (0.3) \\ \searrow \underline{A_2} (0.5) \end{matrix} \}$

Behavior $\{ \underline{s}_1 \rightarrow \begin{matrix} \nearrow A_1 (0.3) \\ \rightarrow \underline{A_2} (0.3) \\ \searrow \underline{A_3} (0.4) \end{matrix} \}$

$\rightarrow B$

$$\boxed{V_{\pi}(s) \stackrel{?}{=} E_{\pi}[G_t | s_t = s]}$$

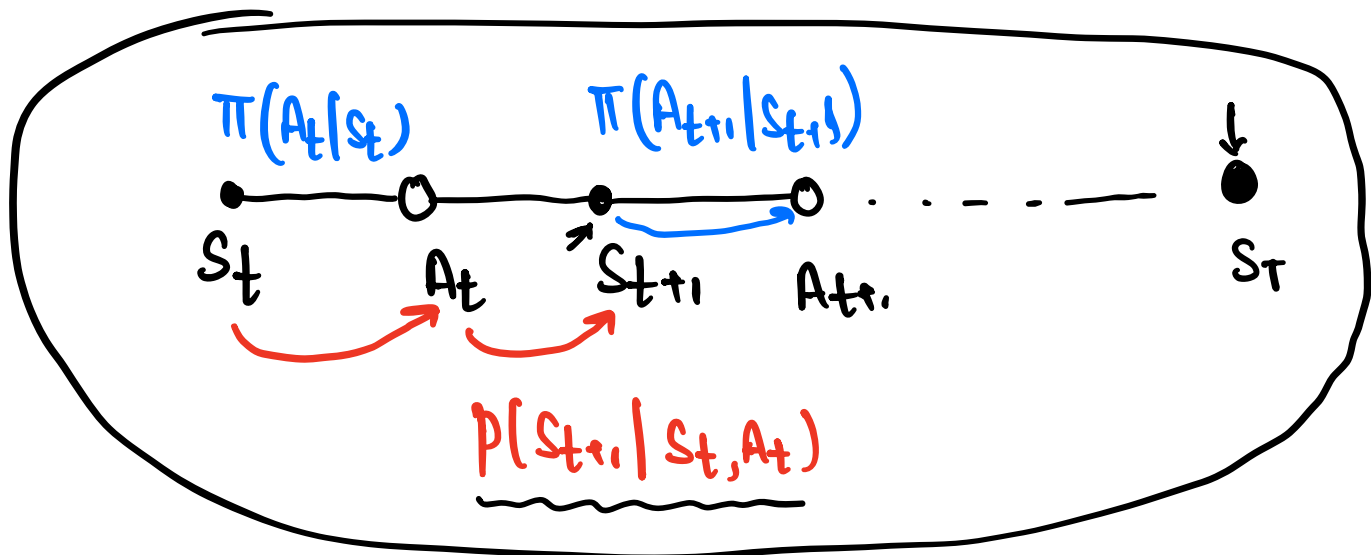
$\uparrow$ $\checkmark$ $\uparrow$
 π

on-policy.

$\rightarrow$ Importance Sampling

Trajectory: $s_t, A_t, s_{t+1}, A_{t+1}, \dots, s_T$
 $\uparrow$
 terminal State

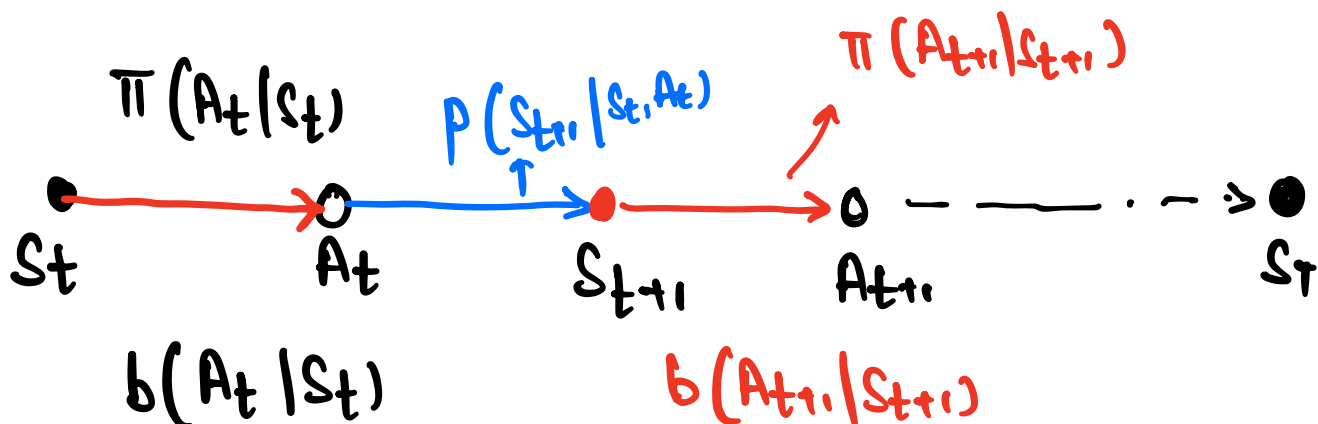
$$\checkmark P\{A_t, S_{t+1}, A_{t+1}, \dots, S_T \mid S_t, A_{t:T-1}, \pi\}$$



$$P\{A_t, S_{t+1}, A_{t+1}, \dots, S_T \mid S_t, A_{t:T-1}, \pi\} =$$

$$\pi(A_t | S_t) P(S_{t+1} | S_t, A_t) \pi(A_{t+1} | S_{t+1}) \dots P(S_T | S_{T-1}, A_{T-1})$$

$$= \prod_{k=t}^{T-1} \pi(A_k | S_k) P(S_{k+1} | S_k, A_k)$$



$$P_{\pi}(\text{Trajectory}) = \prod_{k=t}^{T-1} \pi(A_k | S_k) p(S_{k+1} | S_k, A_k)$$

$$P_b(\text{Trajectory}) = \prod_{k=t}^{T-1} b(A_k | S_k) p(S_{k+1} | S_k, A_k)$$

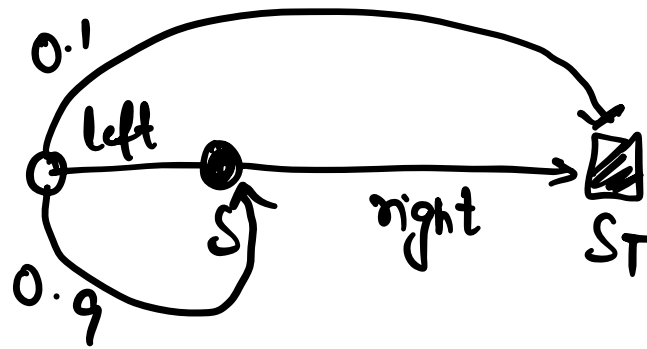
$$\text{IS ratio} = \frac{P_{\pi}}{P_b} = \frac{\prod_{k=t}^{T-1} \pi(A_k | S_k) \cancel{p(S_{k+1} | S_k, A_k)}}{\prod_{k=t}^{T-1} b(A_k | S_k) \cancel{p(S_{k+1} | S_k, A_k)}}$$

$$\text{IS ratio} = \frac{\prod_{k=t}^{T-1} \pi(A_k | S_k)}{\prod_{k=t}^{T-1} b(A_k | S_k)}$$

$$P_{t:T-1}$$

$$V_{\pi}(s) = \mathbb{E} \left\{ \underbrace{P_{t:T-1}}_1 \underbrace{G_t}_{\uparrow^b} \mid S_t = s \right\}$$

$$\underline{V_{\pi}(s)} = E_{\pi} \left\{ \underbrace{G_t}_{\pi} \mid \underbrace{S_t = s}_{\pi} \right\}$$



Target policy: always left action.

$$\Rightarrow \boxed{\pi(\text{left} | s) = 1} \leftarrow$$

$$\pi(\text{right} | s) = 0.$$

Behavior policy:

$$\boxed{\begin{aligned} \pi(\text{left} | s) &= 0.5 \\ \pi(\text{right} | s) &= 0.5 \end{aligned}}$$

E₁:

S, left, S, right, S_T

E₂:

S, right, S_T.

E₃:

S, left, S, left, S_T.

$$\textcircled{E_1}, E_2, E_3, E_4^x$$

$$V(s) = ?$$

$$\underline{V(s_1)}$$

$$\Rightarrow V(s) = \underbrace{p_{0:t:T-1} G_{t1}}_{\textcircled{1}} + \underbrace{p_{1:t:T-1} G_{t2}}_{\textcircled{2}} + \underbrace{p_{2:t:T-1} G_{t3}}_{\textcircled{3}}$$

$$\Rightarrow \textcircled{3}$$

$$\Rightarrow V(s) = \frac{\sum_{t \in \tau(s)} p_{t:T}(t) G_t}{|\tau(s)|}$$

State Value under off policy MC

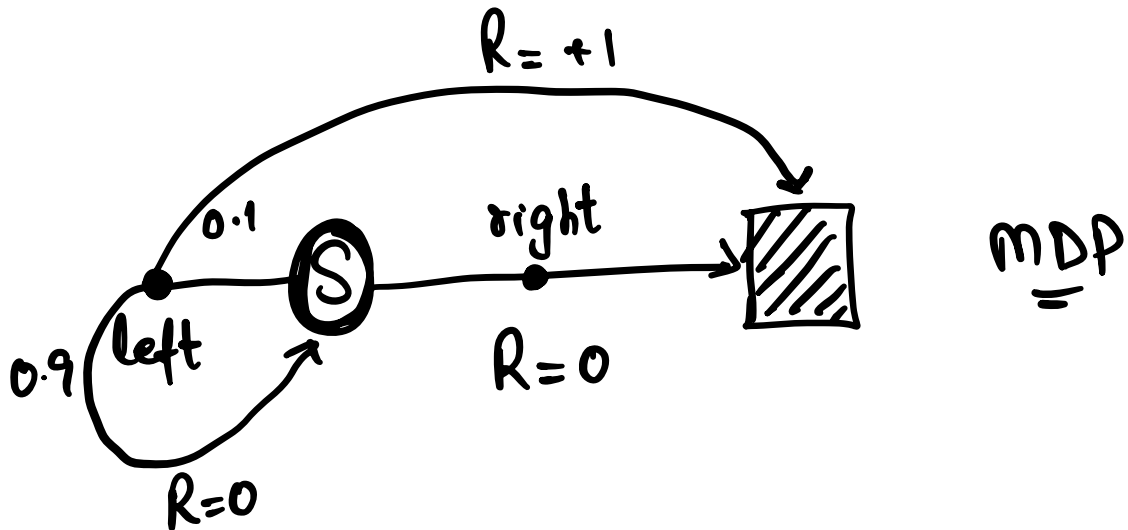
Division by no. of episodes $\rightarrow$ Ordinary IS.

weighted IS:

$$V(s) = \underline{p_{1:t:T-1} G_{t1}} + \underline{p_{2:t:T-1} G_{t2}} + \underline{p_{3:t:T-1} G_{t3}}$$

$$\underline{p_1 t:T-1 + p_2 t:T-1 + p_3 t:T-1}$$

Example:-



Policies

$$\begin{cases} \pi(\text{left} | s) = 1 \\ b(\text{left} | s) = 1/2 \end{cases} \quad b(\text{right} | s) = 1/2$$

MDP
Transition

$$\begin{cases} p(s_T | s, \text{left}) = 0.1 & R = +1 \\ p(s | s, \text{left}) = 0.9 & R = 0 \\ p(s_T | s, \text{right}) = 1 & R = 0. \end{cases}$$

Episode - I :-

S , left, S_T
+1

$$\pi(\text{left} | S) = 1.$$

$$b(\text{left} | S) = 1/2.$$

$$v(S) = ?$$

$$\rightarrow p_I = \frac{\pi(\text{left} | S)}{b(\text{left} | S)} = \frac{1}{1/2} = 2.$$

$$\rightarrow G_I = +1.$$

Episode - II :-

S , left, S , left, terminal
0 +1.

$$p = \frac{\pi(\text{left} | S) \pi(\text{left} | S)}{b(\text{left} | S) b(\text{left} | S)}$$

$$p_{II} = \frac{1}{1/2 \times 1/2} = 4$$

$$G_{II} = 1$$

Episode - III ∴

s, right, ⁰terminal.

$$p = \frac{\pi(\text{right} | s)}{b(\text{right} | s)} = 0$$

$$G = 0.$$

Episode - IV ∴

s, ⁰left, s, ⁰left, s, ⁰left, s, ⁺¹left, S_T

$$p = \frac{[\pi(\text{left} | s)]^4}{[b(\text{left} | s)]^4} = \frac{[1]^4}{[1/2]^4} = 16$$

$$G = 1.$$

Ordinary
II

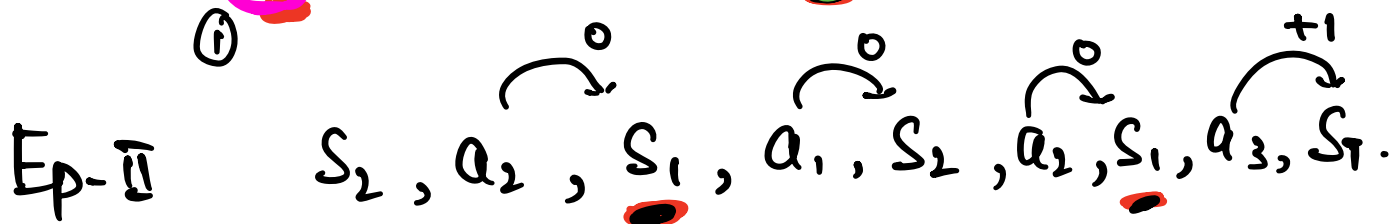
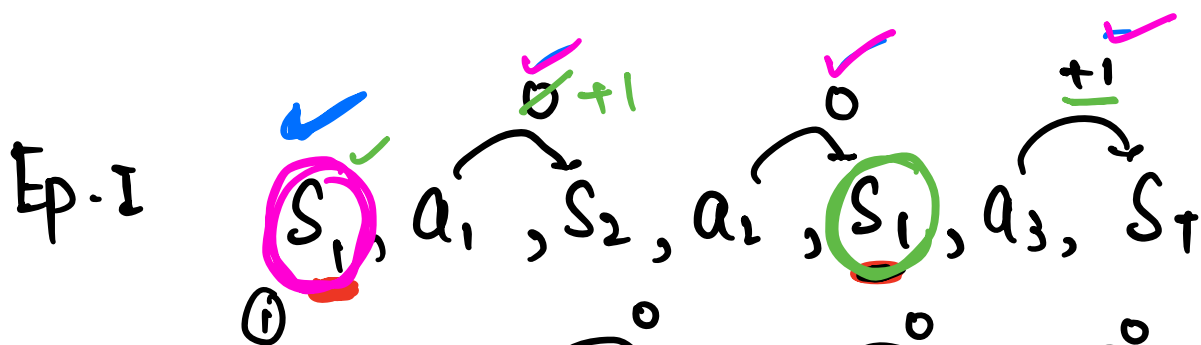
$$V(s) = \frac{\overset{\textcircled{1}}{2 \times 1} + \overset{\textcircled{2}}{4 \times 1} + \overset{\textcircled{3}}{0 \times 0} + \overset{\textcircled{4}}{16 \times 1}}{4}$$

$$= 22/4 = \underline{\underline{5.5}}$$

weighted
C_{IS}

$$V(s) = \frac{\downarrow 2 \times 1 + \downarrow 4 \times 1 + \downarrow 0 \times 0 + \downarrow 16 \times 1}{2 + 4 + 0 + 16}$$

$$= \frac{22}{22} = \underline{\underline{1.0}}$$



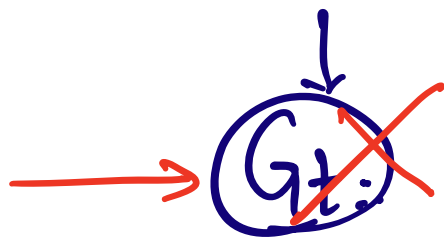
first visit

$$V(s_1) = \frac{1 + 1}{2} = 1.$$

Every visit

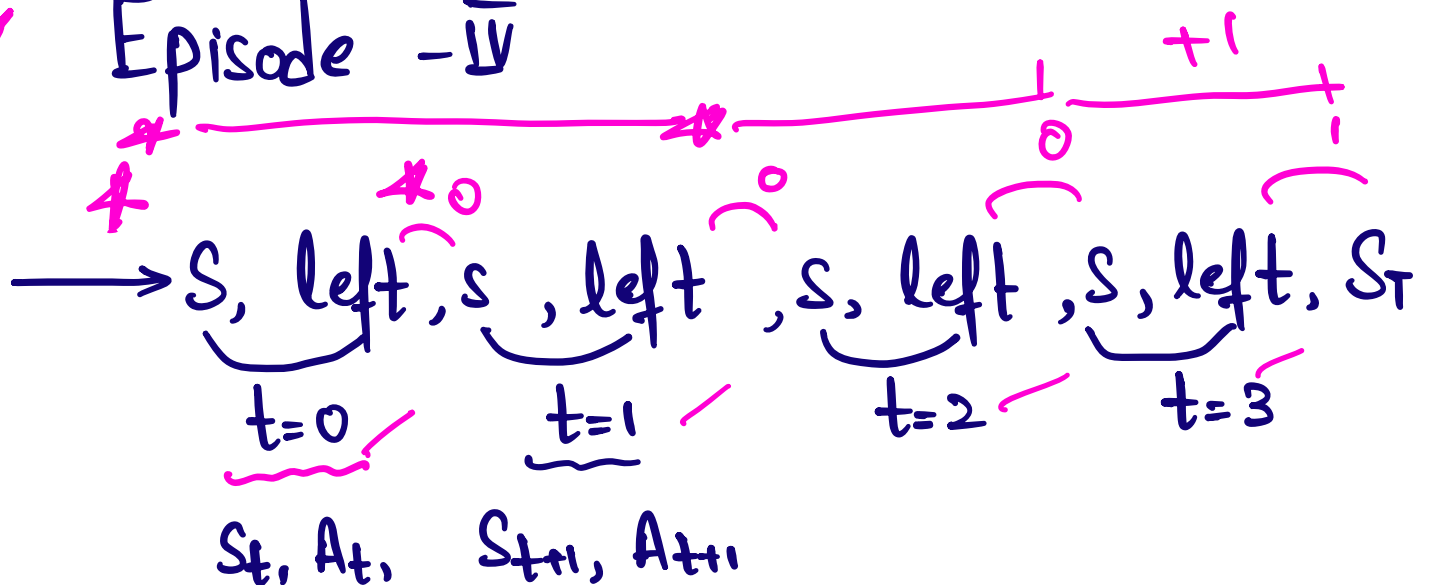
$$V(s_1) = \frac{\checkmark 1 + \checkmark 1 + 1 + 1}{4} = 1.$$

→ Discounting Aware Importance Sampling



→ $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$

★ Episode - IV



Po: t

G_t

$$t=0:-$$

$$\gamma^t = \gamma^0 = 1$$

$$\frac{\pi(l|s)}{b(l|s)} = \frac{1}{\frac{1}{2}} = 2.$$

$$\gamma = 0.9$$

$$t=0:- \gamma^t = \gamma^0 = 1.$$

$$P_{0:t} \frac{\pi(l|s)}{b(l|s)} = \frac{1}{\frac{1}{2}} = 2.$$

$$G_t$$

$$\frac{1}{0}$$

$$\times \frac{(1-\gamma) \gamma^t P_{0:t}}{(1-0.9)(1)(2)} = 0.1 \times 2 = 0.2.$$

$$t=1:- \gamma^1 = (0.9)^1 = 0.9$$

$$P_{0:t} \frac{\pi(l|s) \pi(l|s)}{b(l|s) b(l|s)} = 4$$

$$1$$

$$(1-0.9)(0.9)(4) = 0.36.$$

$$t=2:- \gamma^2 = (0.9)^2 = 0.81$$

$$P_{0:t} 8$$

$$\frac{1}{1}$$

$$(1-0.9)(0.81)(8) = 0.648.$$

$$t=3:- \gamma^3 = (0.9)^3 = 0.729$$

$$P_{0:t} 16$$

$$1$$

$$\gamma^t P_{0:t} G_t = 0.729 \times 16 \times 1 = 11.664.$$

$$E_{p-\bar{IV}} :- \frac{0}{2} + \frac{0}{36} + \frac{0.648}{11.664} + 11.664 = 12.87$$

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ms(AI) E-13

Bonus.



3-marks

Episode - II.

$s, \text{left}, s, \text{left}, \text{terminal.}$
 ~~$t=0$~~ $t=0$ ~~$t=1$~~ $t=1$ $+1$

$t=0$ γ^t $\rho_{0:t=0}$ $G_{0:t=0}$
 $\gamma^0 = 1$ $\pi(l|s) = \frac{1}{2} = 2.$ 0
 $b(l|s) \frac{1}{2}$

$t=1$ γ^t $\rho_{0:t=1}$ $G_{0:t=1}$
 $\gamma^1 = (0.9)$ $\pi(l|s)\pi(l|s) = 4$ $+1.$
 $= 0.9$ $b(l|s)b(l|s)$

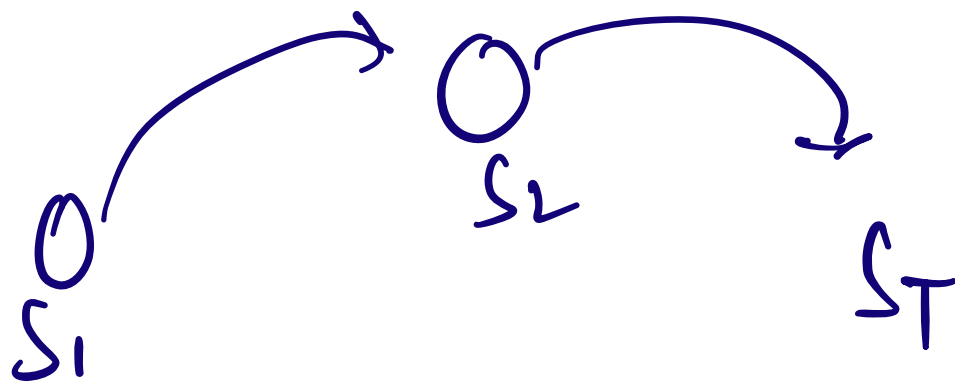
$G_t = \text{Ep-II} = 0 + (0.9)(4)(1)$
 $= \cancel{0.9} 3.6$

$$\frac{- + 3.6 + 0 + 11.664}{4}$$

$$V_{\pi}(s) = \frac{- + 3.6 + 0 + 11.664}{4}$$



→ MDP



→ Episodes:.

o) true state values: ✓

1 iteration

← i) FVMC

← 2) EV MC

← 3) TD ✗

← 4) SARSA ✗

← 5) Q-learning ✗

← 6) IS (ordi / weighted)

← 7) Discounting aware Is.

⇒ MC & TD

(~~bias~~ implementation difference).

(bias & variance diff)

(bootstrap).

MSE = (bias)² + variance.

→ off policy & on-policy (diff)
↳ Assumption of coverage